THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · EASY LEVEL CHAPTER TEST SET 3

Chapter 11: **Areas Related to Circles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A sector of a circle of radius 21 cm has central angle 120°. The area of the sector is: 1

(a) 462 cm² **(b)** 44 cm² **(c)** 1386 cm² **(d)** 924 cm²

2. A sector of a circle of radius 21 cm has central angle 120°. The length of its arc is: 1

(a) 86 cm **(b)** 462 cm **(c)** 44 cm **(d)** 88 cm

3. A semicircular protractor-shaped plate has radius 21 cm. The perimeter of the plate is: 1

(a) 174 cm **(b)** 66 cm **(c)** 108 cm **(d)** 87 cm

4. A quadrant of a circle has radius 21 cm. The perimeter of the quadrant is: 1

(a) 33 cm **(b)** 75 cm **(c)** 108 cm **(d)** 54 cm

5. The radius of a circle is 28 cm. The area of the circle is: 1

(a) 2464 cm² **(b)** 4928 cm² **(c)** 176 cm² **(d)** 88 cm²

6. The radius of a circle is 35 cm. The circumference of the circle is: 1

(a) 3850 cm **(b)** 220 cm **(c)** 110 cm **(d)** 440 cm

7. The circumference of a circle is 176 cm. The radius of the circle is: 1

(a) 56 cm **(b)** 88 cm **(c)** 28 cm **(d)** 14 cm

8. The area of a circle is 2464 cm^2. The radius of the circle is: 1

(a) 392 cm **(b)** 784 cm **(c)** 56 cm **(d)** 28 cm

9. A sector of a circle of radius 21 cm has central angle 180°. The area of the sector is: 1

(a) 1386 cm² **(b)** 693 cm² **(c)** 2079 cm² **(d)** 66 cm²

10. A sector of a circle of radius 21 cm has central angle 180°. The length of its arc is: 1

(a) 693 cm **(b)** 66 cm **(c)** 108 cm **(d)** 132 cm

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A sector of a circle of radius 21 cm has central angle 150°. Find the area of the sector. 2

12. The radius of a circle is 24.5 m. Find the area of the circle. 2

13. The radius of a circle is 24.5 m. Find the circumference of the circle. 2

14. A sector of a circle of radius 21 cm has central angle 40°. Find the area of the sector. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A sector of a circle of radius 21 cm has central angle $36°$. Find the length of its arc. 3

16. A sector of a circle of radius 21 cm has central angle $180°$. Find the perimeter of the sector. 3

17. A semicircle has radius 35 cm. Find the area of the semicircle. 3

OR

A sector of a circle of radius 21 cm has central angle $24°$. Find the length of its arc. 3

18. A sector of a circle of radius 21 cm has central angle $150°$. Find the perimeter of the sector. 3

OR

A semicircle has radius 42 cm. Find the area of the semicircle. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A circular park of radius 14 m is surrounded by a path 14 m wide. Find the area of the path. 5

OR

The largest possible circle is cut from a square sheet of side 35 cm. Find the area of the sheet left over. 5

20. A circular park of radius 7 m is surrounded by a path 7 m wide. Find the area of the path. 5

OR

The largest possible circle is cut from a square sheet of side 7 cm. Find the area of the sheet left over. 5

END OF QUESTION PAPER — Total: 40 Marks

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